



MECH MAINT- TURBINE Department
SIMHADRI

Note No :1



Ref:SMTTP/MECH MAINT-/2023-24/LMI FRF/487896

Date:

15/09/2023(13:23:03)

Subject:Revision of LMI “The examination, Maintenance, safety and handling aspects of phosphate ester fire resistant hydraulic fluids

CC-OS has revised the OGN with ref.no: OGN/OPS/MECH/001 vide Rev. No.: 4 dated Aug. 2022

So, the LMI: **“THE EXAMINATION, MAINTENANCE, SAFETY AND HANDLING ASPECTS OF PHOSPHATE ESTER FIRE RESISTANT HYDRAULIC FLUIDS”** with LMI NO.: **LMI/OGN/OPS/MECH/001** is due for revision and necessary changes are done in the LMI.

The details of the changes are mentioned in the revision index and are as follows:

A] Changes done in following sections

1. Examination of New Fluid on Delivery
2. Significance of Monitoring Parameters
3. Purification Equipment
4. Avoidance of mixing natural and synthetic-based fluids within the same control system

B] Responsibility of carrying out different activities w.r.t. oil replacement/top-up/ schedule of replacement of filter elements, testing of oil samples through OEM/NETRA have been incorporated in this document.

C] Following two TC Documents merged together:

1. OGN/OPS/MECH/001 Rev 03, June-2020: The Examination, Maintenance, Safety & Handling Aspects of Phosphate Ester Fire Resistant Hydraulic Fluids
2. OGN/OPS/MECH/014 Rev 01, June-2020: Fire Resistant Fluids for Control Systems-Avoidance of Mixing Natural and Synthetic Based Fluids within the same Control System

Complete LMI is revised as revision no. 6 inline with revised “OGN/OPS/MECH/001 Rev. No.: 4, dated Aug. 2022”

Revised LMI as revision no. 6 is submitted for Approval of Competent authority.

Enclosures:

1. OGN/OPS/MECH/001 Rev. No.: 4 Aug. 2022
2. Revised LMI (Revision: 06)
3. Earlier approved LMI (Revision: 05)

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LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/001/

ISSUE NO.1; REVISION NO. 6.00; DATE: SEPT'2023

**THE EXAMINATION,
MAINTENANCE, SAFETY AND
HANDLING ASPECTS OF
PHOSPHATE ESTER FIRE
RESISTANT HYDRAULIC FLUIDS**

APPROVED FOR

***Implementation*-----**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	THE EXAMINATION, MAINTENANCE, SAFETY AND HANDLING ASPECTS OF PHOSPHATE ESTER FIRE RESISTANT HYDRAULIC FLUIDS
<u>2.</u> a) <u>DOCUMENT CODE</u> b) REVISION/ISSUE NO.&DATE	<u>LMI/OGN/OPS/MECH/001/</u> <u>ISSUE-1; REV.-6.00; SEPT'2023</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>OGN/OPS/MECH/001</u>
<u>4.PRERPARED BY</u>	SR. MANAGER(TMD)
<u>5. CHECKED BY</u>	HOD (TMD) HOD (CHEM)
<u>6.RECHECKED BY</u>	GM(MAINT)
7. REVIEWED BY	GM(O&M)
<u>8. APPROVED BY</u>	HOP
<u>9.DATE OF ISSUE</u>	
<u>10.INDEX LIST</u>	ENCLOSED
<u>11. DISTRIBUTION LIST</u>	ENCLOSED

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TITLE: THE EXAMINATION, MAINTENANCE, SAFETY AND HANDLING ASPECTS OF PHOSPHATE ESTER FIRE RESISTANT HYDRAULIC FLUIDS		
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<u>2.0 INTRODUCTION</u> The hydraulic lines of the electro-hydraulic control system of turbine are often near the high temperature steam lines. To reduce the hazard of fluid ignition should the fluid leakages come in contact with metal surfaces a fire-resistant fluid (FRF) based on 100% Natural trixylenly phosphate (TXP). The FRF for Simhadri as originally provided by OEM is procured under a brand name of "FYRQUEL EHC-N". Phosphate ester fluids differ from petroleum mineral oils in respect of several important properties which necessitate special examination, maintenance, safety precautions & handling procedures. The purity of fire-resistant fluid is of vital importance to the reliability of the plant. At Simhadri the control system is without servo valves which means a normal category of duty for FRF fluids. The objective of this LMI is to regiment a framework for monitoring and maintaining the performance characteristics and safe handling and disposal of fire-resistant fluids. <u>3.0 REFERENCE DOCUMENT</u> Operation guidance note OGN/OPS/MECH/001 Rev. No.: 4 Aug. 2022 circulated by operation services-NTPC corporate center. <u>4.0 SUPERSEDED DOCUMENT</u> LMI/OGN/OPS/MECH/001, ISSUE NO: 1, REVISION NO: 5, DATE: JULY'2022 <u>5.0 SCOPE</u> Part I of this Guidance note LMI is intended to help chemists and operational staff to appreciate the important properties in the control systems of such diverse equipment as turbines, boiler dampers, feed water and steam valves and feed and cooling water pumps. It also specifies the requirements for examining consignments of new fluid and for monitoring and maintaining adequate fluid condition. Part II is a guide to the safe handling, storage, use and disposal of phosphate esters. Changes are made from time to time to the fluid specification itself to take account of new equipment and operational requirements and of fluid developments.		

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6.0 PART-I: EXAMINATION AND MAINTENANCE ASPECTS

6.1 Fluid Properties

There are many designs (types) of hydraulic equipment each of which subjects the fluid to conditions which affect its performance and properties. The most stringent conditions exist in equipment operating at high pressure (up to approximately 140 bar) where pump and valve clearance are of the order of 2 to 4 μm . The passage of fluid through such gaps imposes a high rate of shear, which produces local high temperatures accompanied by fluid degradation. The mechanical and chemical erosive properties of the fluid, of its degradation products and particulate contaminants are then of great importance. Fluids have to be chosen, therefore, which will withstand these conditions whilst equipment has to be designed to be compatible with and maintain, a satisfactory fluid condition. The properties which have the greatest influence on equipment design and operation are:

- (i) Viscosity
- (ii) Density
- (iii) Thermal and oxidation stability
- (iv) Air release
- (v) Foaming
- (vi) Resistivity (corrosion-erosion)
- (vii) Cleanliness

For equipment subject to water ingress:

- (i) Hydrolytic Stability
- (ii) Anti-rust (anti-corrosion)
- (iii) Demulsibility

Phosphate ester fluids are particularly susceptible to thermal and hydrolytic degradation and to a lesser extent to oxidation. Temperature and water are, therefore, of particular importance in all types of equipment and must be controlled. The products of degradation particularly in the presence of water are corrosive and can readily erode finely machined parts. Particulate contaminants in fluid moving under high velocity conditions, whether they are present in the original fluid or are produced by wear and corrosion in the system, can also be highly erosive. In order to maintain the fluid in a satisfactory condition, line filters are necessary for all equipment and chemical absorption filters for most equipment. Additionally, desiccant breathers and in some cases vacuum dehydrators are installed to minimize water contamination. To be reasonably sure that these are functioning correctly and that the fluid is maintained in a satisfactory condition it is necessary to monitor selected properties of the fluid. In order to minimize the number of tests carried out equipment has been divided into three categories, namely very severe, severe and normal, depending upon its design, function, location and fluid performance.

Air is normally entrained in the drain pipe work of a system and the fluid reservoir permits some detrainment because of the transit time of the fluid through the tank. The air content of the fluid at the reservoir outlet passing to the main fluid pump when the pump is supplying its normal flow rate through an operating system is most meaningful and should be checked during commissioning.

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<u>6.2 Examination of New Fluid on Delivery</u> It is essential that all incoming supplies of phosphate ester fluid be adequately monitored to guard against incorrect or contaminated material being delivered. Fluid intended for filling or additions to very severe systems must be added via a suitable fine filter. Experience has shown the need for common procedures to be used for the sampling, examination and acceptance of incoming supplies of phosphate ester fluids. It is essential that personnel responsible for sampling and testing shall have the necessary experience and skills, and that scrupulous attention to details be applied at all times to avoid erroneous results. Sample is to be collected from any one of the oil drums of a particular lot. All samples should be immediately examined for appearance. The tests in Table 2 should be completed on the bulk sample before the phosphate ester fluid is used in service. Individual samples should be retained until the bulk sample is passed as satisfactory. Where post-testing is undertaken an additional sample must be available and reserved for future analysis by the manufacturer or an alternative laboratory to that of the receiving station. The specification test limits are given in Table 3. If fluids not complying with the specification are received at any location, the Supplier, O&M in-charge, and the materials stores should be informed. In addition, for commissioning stations, the Commissioning in-charge and Plant Manufacturer should be informed. <u>6.3 Deterioration and Important Properties of Fluids in Service</u> The Table 1 lists the parameters which require monitoring for the three categories of duty for phosphate ester fluids together with other parameters of interest. Limiting values are assigned to each parameter as well as the frequency of monitoring. The monitoring frequency suggested is one, which will enable the detection of any slow deterioration of the system during normal operation. During commissioning, and following observation of abnormal condition, an increased monitoring frequency may be indicated. In the long term a parameter such as D.C resistivity may be sensitive to small changes in acidity and moisture content. The purposes of monitoring are: i. To follow the progress of system flushing operations on the installation of new plant to ensure that this is properly carried out. ii. To identify faulty items of plant in advance of plant breakdown. iii. To identify faulty maintenance operations before the effects become serious. iv. To suggest what corrective action is necessary if a parameter has reached its limiting value. v. To ensure that the operating conditions are satisfactory for sensitive control elements e.g. servo valves. vi. To establish when the useful life of the fluid has elapsed. There is a need to consult specialists in fluids or plant items concerned, if the reasons for an off-limit value cannot be determined.		

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6.4 Significance of Monitoring Parameters					
Table 1					
S.N.	Parameter (trend)	Cause of change	Harmful Effects	Remarks	
1	Acidity (increase)	- Degradation of fluid due to increase in moisture content or high fluid temperatures. - Chemical contamination of system - Ineffective operation of Fullers Earth filters where these are fitted.	Corrosion or equipment malfunction if allowed to persist, particularly in the presence of moisture.	Fortnightly monitoring	
2	Moisture Content (increase)	- Abnormal ingress of water, e.g. seal failure, faulty top-up procedure, ineffective reservoir breather, cooler fault. - Normal ingress of water, but inadequacy of dehydration equipment. - The density of water is less than that of phosphate ester and it will therefore float on top making flushing out of systems containing water contamination difficult.	Long-term system and fluid deterioration, corrosion, rusting and seizure of sensitive control elements.	Fortnightly monitoring	
3	Particulate contamination (increase)	- Onset mechanical failure of a plant item (Self-generation of particles) - Inadequate maintenance - High Fluid temperatures - Failure of seal permitting ingress of contamination or failure of a filter element	- Valve silting and malfunction - Particulate erosion giving rise to a loss of performance.	Monthly monitoring	
4	DC Resistivity (decrease)	- Increase in acidity and/or moisture content - Topping-up with an unsuitable grade of fluid - Contamination with electrically conducting material.	- As for SI nos 1 and 2 - Erosion by electrochemical effects giving rise to a loss of performance.	To be tested six-monthly from CC-NETRA	

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5	Mineral Oil Content	- Topping-up with wrong fluid - Residual contamination from components tests in mineral oil. - Failure of common seal with plant items using separate mineral oil lubrication.	- Deposits giving rise to seizure of sensitive control elements. - Decrease of fire resistance.	To be tested six-monthly from CC-NETRA	
6	Viscosity	- Continuous shearing action of pumps on fluid which contain (viscosity Index) additive. - Serious fluid degradation.	- Change in system flow requirements - Failure of system components (especially pumps) operating outside specified viscosity range.	Monthly monitoring	
7	Chlorine Content	Due to contamination by sea water, chlorinated solvent, Chlorine containing protective used as treatment prior to assembly.	- Erosion/corrosion giving rise to a loss of performance - Gross contamination may indicate need to change fluid.	To be tested from external labs during commissioning or abnormal operation (if required)	
8	Aeration	- Abrupt pressure reduction in the presence of air, faulty seals, poor pump suction design. - Increase in air release value due to contamination or chemical changes.	- Pitting damage to pumps and system components where fluid pressure changes occur. - Spongy operation of system.	To be tested from external labs during commissioning or abnormal operation (if required)	
9	Fluid Temperature (bulk fluid temperatures of 60°C and hotspot temperatures of 112°C are compatible with long fluid life).	Persisting high bulk fluid or hotspot temperatures.	Permanent damage to fluid.	To be tested from external labs during commissioning to confirm the design.	

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6.5 Purification Equipment

Depending on the needs of the system design, equipment were provided for the extraction of acidity, water and particulates from the operating system. Such equipment may comprise any/or all of the following items:

- i. Solids Filters
- ii. Fullers Earth filters for the extraction of acidity, sediment and water
- iii. Dehydration equipment
- iv. Desiccant breathers
- v. Vacuum dehydrators
- vi. Blowers on tanks

Fullers Earth filters are provided for acidity absorption, whereas cartridge filters keep a check on mechanical impurities. The above equipment's are fitted in the main system flow path or in a by-pass loop. The duty may be either continuous or intermittent.

The purification equipment plays an important part in maintaining correct system operating conditions and should itself be adequately maintained and periodically checked for correct operation.

The provision of filters to maintain fluid within the requirements of Table 4 should ensure satisfactory fluid life and thus reduce waste arising. To achieve this, earth filters should be replaced when the pressure drop across the filter exceeds the specified value or when the acidity or resistivity of the fluid cannot be controlled within the action level (see Table 1 Sr. No.4).

Table 2: Recommended properties of new batches of FRF (Typical values for Fyrquel EHC-N)

S.N.	Parameter	Allowable Limits	Remarks
1	Water Content (ppm)	Max 1000	To be tested in new oil
2	Kinematic Viscosity cSt at 40 °C (ISO VG 46)	41.4-50.6	
3	Density at 15 °C (kg/M3)	Max 1250	
4	Pour Point (°F)	Max (-18)	
5	Acid Number (mg KOH/g)	Max 0.1	To be tested in new oil
6	Flash Point (°C)	>235	
7	Electrical Resistivity (MΩm)	>50	
8	Ignition Temperature (°C)	>550	
9	Foaming tendency (ml) at 25 °C	Max 100	
10	Air release at 50 °C	Max 3	
11	Wick Flame persistence time (sec)	Max 5	
12	Water Separability (sec)	Max 3000	
13	Demulsification (minutes)	Max 20	
14	Particle Distribution (ISO 4406 code)	Max 15/12	
15	Chlorine Content (mg/kg)	Max 50	

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Table 3: Tests and frequency for Phosphate ester -Fire Resistant Fluid: Responsibility-HOD (Chemistry)								
S.N.	Parameter	FRF De-gradation level	Frequency of Testing	Allowable Limits	Remarks			
1a	Acidity	Normal (< 0.1)	Fortnightly	Max 0.1 mg KOH/goil	Limits of parameters may change based on OEM recommendations.			
1b		Bad (0.1~ 0.2)	Weekly					
1c		Severe/ Very severe (>0.2)	Frequency to be increased					
2a	Moisture content	Normal (< 1000)	Fortnightly	Max 1000 ppm				
2b		Severe (1000-1500)	Weekly					
2c		Very severe (>1500)	Frequency to be increased					
2a	Particulate contamination	Normal (Upto NAS 7)	Monthly	Max 7 (NAS 1638) or 18/16/13 (ISO 4406)				
2b		Bad (NAS 7-9)	Fortnightly					
2c		Severe (NAS >9)	Weekly (or more frequent if required)					
Note: Oil samples are to be checked six-monthly from CC-NETRA or IOCL/supplier for all the above parameters and additional parameters viz. Color, Density, Foaming Characteristics, DC resistivity, *mineral oil content (*on special circumstances if required by site).								
Table 4: Schedule of Replacement of Filter cartridges -Provided in Fire Resistant Fluid Re-Generation circuit: Responsibility-HOD (TMD)								
S.N.	Parameter	Allowable Limits	FRF De-gradation level	Replacement of Cartridge	Replacement Schedule			
1a	Acidity	Max 0.1 mg KOH/g oil	Normal (< 0.1)	Fullers earth cartridge	Once in a year			
1b			Bad (0.1~ 0.2)		When Re-regenerative system inlet & outlet acidity level saturates or Max 6 monthly intervals			
1c			Severe (0.2~ 0.3)		When Re-regenerative system inlet & outlet acidity level saturates or Max 3 monthly intervals			
1d			Very Severe (>0.3)		Replacement of Fullers earth cartridges with Strong / weak base anion resin as per NETRA recommendations.**			

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2a	Moisture content	Max 1000 ppm	Normal (<1000)	Dry filter cartridge	Once in a year
2b			Sever (1000-1500)		When Re-generative system inlet & outlet moisture level saturates or Max 3 monthly intervals
2c			Very Severe (>1500)		Use of portable Vacuum De- Hydration machine
2a	Particulate contamination	Max 7 (NAS 1638) or 18/16/13 (ISO 4406)	Normal (Upto NAS 7)	Fine filter cartridges	Once in a year
2b			Bad (NAS 7-9)		Six-Monthly
2c			Severe (NAS >9)		Six-Monthly

Note:

(i) ** Vacuum de-hydration system should be kept into service before use of strong/weak base anion instead of fuller's earth, to take care of increase in moisture in FRF.

(ii) Use of external mobile filtration machine with filter element up to one-micron size should be kept in service continuously in each unit.

7.0 PART-II: SAFETY AND HANDLING ASPECTS

7.1 Description:

The use of fire-resistant fluids of the phosphate ester type in hydraulic control systems is becoming more wide spread. These fluids may readily be confused with mineral oils, and it is necessary to draw attention to the need for additional precautions to be taken in their storage, handling, use and disposal.

The most serious risk is that of swallowing the fluids, a lesser hazard is associated with eye and skin contamination and with inhalation of fumes.

All recommendations regarding safety should be considered in conjunction with relevant ISS code.

It is recommended that the following actions should be taken whenever such fluids are in use.

7.2 Administrative Actions

7.2.1 It should be ensured that all staff is aware of the nature of the material they are handling

and know the rules set out below.

7.2.2 To avoid confusion, fire resistant fluids must be clearly labelled in stores and kept separate from mineral oils, which may be used for similar purposes. All purchased fluids must have appropriate hazard data sheets.

7.2.3 The disposal of waste materials must be carefully controlled, particularly regarding the need to avoid contamination of water courses and agricultural land.

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<u>7.3 Handling Precautions</u> 7.3.1 All containers of fire-resistant hydraulic fluid must be clearly marked. Fluid which has leaked should be collected in clearly marked containers. 7.3.2 Eating, drinking or smoking must be prohibited whilst the fluid is being handled, to prevent contamination of the lips and mouth and minimize risk of ingestion. 7.3.3 Splashes may cause severe irritation and goggles must be worn at all times when the fluid is being handled or when working on hydraulic control or other systems containing the fluid. If an eye splash does occur, the affected eyes should be irrigated with water as soon as possible and the casualty referred to the First Aider for appropriate treatment. 7.3.4 To avoid contact of the fluid with the skin, suitable impermeable protective gloves must be worn. Any contaminated clothing must be washed as soon as possible. Where under garments are contaminated a shower must be taken. After handling the fluid, and before eating, drinking or smoking, the hands must be washed thoroughly. 7.3.5 If leakage of fluid occurs on to hot pipe work, white choking fumes containing phosphorous oxide may be formed which can cause irritation of the throat and chest. Under such circumstances an emergency procedure must be set up and the fault corrected as soon as possible. Breathing apparatus must be worn by anyone working in an area exposed to the fumes. 7.3.6 Spillage must be prevented, as far as possible, from entering surface drainage channels. 7.3.7 All waste fluids must be considered potentially toxic and be disposed through an authorized waste disposal organization. A preferred arrangement would be to return the fluid to the supplier for reprocessing and this will be the future aim of national discussions with the supplier. Note :- M/s Stauffer France have suggested use of lime for neutralization of Fyrquel. They have also indicated that waste Fyrquel can be sold to some plastic industry in India where it can be utilized as elastomer for making the plastic soft. For disposal it can be shipped back to the supplier or buried in sealed drums inside or under covered RCC Tanks. <u>7.4 First Aid</u> In the event of first aid being necessary, advice should be sent from the hospital. 7.4.1 If the fluid is swallowed, do not induce vomiting. The casualty must be referred to hospital with a note (see 7.4.5) ensuring that the hospital casualty department is informed that phosphate ester has been swallowed. 7.4.2 If significant inhalation of fumes has taken place, the casualty must be kept at rest and referred to hospital with a note (see 7.4.5) 7.4.3 If the material enters the eye, the affected eye must be thoroughly irrigated for 15 minutes using clean tap water. The casualty should then be referred for a medical opinion with a note (See 7.4.6). 7.4.4 If Signs of skin irritation occur refer for a medical opinion. 7.4.5 If hospital reference is necessary as in 7.4.1 and 7.4.2 above, a pro-forma note as given in Appendix- 1A must be completed and sent with the casualty. 7.4.6 If hospital reference is necessary as in 7.4.3 above, a pro-forma note as given in Appendix-1B must be completed and sent with the casualty.		

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8.0 AVOIDANCE OF MIXING NATURAL AND SYNTHETIC-BASED FLUIDS WITHIN THE SAME CONTROL SYSTEM

8.1 OPERATION REQUIREMENTS

Because of concern over the effects of mixing of the two types of fluid in the same system, users of FRFs are advised not to mix natural and synthetic before the results of any field trials are known. This is in spite of any apparent compatibility which may be suggested in the product data.

If oil companies cannot supply the currently used natural FRF in sufficient quantities for topping up purposes, it will be necessary to change the full charge of fluid until such time as the results of any mixing trials are known.

Fluids from different suppliers, which may be derived from the same source but have different additives, should not be mixed without prior consultation with Engineering Division and continuity of condition monitoring by the original supplier.

8.2 RECOMMENDATIONS

- Natural and synthetic based FRFs must not be mixed within the same system.
- If supplies of the previously used FRF, from either base, cannot be obtained for topping up purposes, then the full system charge must be changed to the fluid of the revised base.
- FRFs used for lubrication purposes may have hydrocarbon oils and additives of different quality. So, they should not be mixed.

RESPONSIBILITY:

- Chemistry- for quality checks of oil for replacement purpose.
- Operation & TMD for oil top-up/replacement.

9.0 EXCEPTION REPORTING

Any exceptions to the instructions specified in the LMI are to be reported as per the following format to HOD (MM)

Unit No.	Description of exception	Remarks

Responsibility- HOD (TMD) & HOD (CHEMISTRY)

10.0 RESPONSIBILITIES

The responsibility of carrying out different tasks as detailed in the LMI are with HOD (TMD) & HOD (CHEMISTRY)

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11.0 RECORDS TO BE MAINTAINED

Sl. No.	Record Description	Maintained by	Frequency	Remarks
A. For routine FRF analysis:				
1	Test results of routine analysis of FRF samples	Chemistry	As per table 3	
2	External test reports of CC-NETRA	Chemistry (external test report copies to be available TMD)	Six-Monthly or earlier if required	
3	External test reports of IOCL/Supplier	Turbine Maintenance (external test report copies to be available with Chemistry)	Six-Monthly or earlier if required	
B. For Oil replacement/Top-up:				
1	Records must be kept for all inspections and should contain the following data: - Details of oil top-up - Details of oil replacement & reasons - Quality checks of oil for replacement	Operation & TMD Operation & TMD Chemistry	As per requirement	
C. Replacement of filter cartridges:				
1	Replacement of Filter cartridges as per table 4	Turbine Maintenance	As per Schedule (in table 4)	

11.REVIEW

This document shall be reviewed by HOP-SIMHADRI on two-yearly basis, or as and when required.

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REVISION INDEX (for changes/revision)					
Sl. No.	Rev. no.	Date	Details	Approved by	
1	2.00	APR'2013	Point No-5: Revision number & Date changed Point No-7: Allowable limits changed for acidity & kinematic viscosity Point No-7.0-I(c): Deleted Point No-7.0-III(g): changed Point No-7.0-II(d): Measure for restoration revised for maintaining moisture < 1000 ppm Point no-7,8,9: Responsibility changed from DGM (TMD) to HOD (TMD); DGM (Chemistry) to HOD (Chemistry) and AGM (MAINT) to HOD (MM). Distribution list: changed. Annexure-I & II: Material contains changed.	HOP	
2	3.00	JAN'20	Point no-11: Review period changed to two yearly basis from three-yearly basis. Point no-5: Revision no and date changed Distribution list: changed.	HOP	
3	4.00	JULY'20	Point no-7 I b: Table: Method of testing, Test result limits and frequency values changed Point no: 7 I c: Sampling procedure for new drums added Point no: 7 II d: DC resistivity & Mineral oil content rows added in the table Point no: 7 III g to i: Disposal of oil clause added Point no: 7 III: First Aid: First aid portion made separate.	HOP	
4	5.00	JULY'22	Point no-7 I b: Table: Moisture content & acidity Test result limits values changed	HOP	
5	6.0	SEPT'23	Complete LMI revised inline with revised OGN “OGN/OPS/MECH/001 Rev. No.: 4 Aug. 2022”	HOP	

[illegible]

NTPC	COMPLIANCE RECORD		SIMHADRI
TITLE: THE EXAMINATION, MAINTENANCE, SAFETY AND HANDLING ASPECTS OF PHOSPHATE ESTER FIRE RESISTANT HYDRAULIC FLUIDS REV: 06 DATE: SEPT'2023			
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UNIT NO.	ACTIVITY	COMPLIANCE	

NTPC	ANNEXURE-1A	SIMHADRI
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Pro-forma to be completed and sent to the hospital with a casualty who has swallowed phosphate ester fluid or inhaled significant quantities of phosphate ester fumes.

To

Casualty Officer

Date.....

Name :

Address :

Station :

The above-mentioned has swallowed/inhaled* a quantity of @

This material contains tri-aryl phosphate.

The following first aid treatment has been carried out

Signed:

Note : The Medical Advisor can be contacted at
 Tel Nowith regard to any possible long-term effects.

*Delete as necessary

@ Insert proprietary name of material

NTPC	ANNEXURE-1B	SIMHADRI
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Pro-forma to be completed and sent to the hospital with a casualty who has received eye contamination with phosphate ester fluid

To:

Casualty Officer

Date

Name :

Address :

Station

The above-mentioned has been splashed in the eye with+

The material contains tri-aryl phosphate

The following first aid treatment has been carried out :

.....
.....
.....

In accordance with the recommendation of the Medical Advisor, whose notes on these conditions are below, he is referred to you for further treatment.

Signed :

Notes on Effects and Treatment of Splashes from Fire Resistant Fluids.

These fluids are all based on similar chemicals, which may decrease the skin and which are extremely painful in the eye. They seldom produce any permanent damage. Splashes should be washed off immediately.

In the case of the eye, it should be irrigated for 15 minutes.

The case should then be referred to a doctor for local anesthesia to relieve pain.

The eye should then be covered until it returns to normal.

.....
Medical Advisor

+Insert proprietary name of material

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SAFETY DATA SHEET

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Product id	7071E	
Revision date	20/10/2013	Revision: 5
Supersedes	21/02/2012	

1. Identification of the substance & the company

Chemical name	Trixylyl phosphate
Synonym(s)	TXP; Phenol, dimethylphosphate (3:1). Trixylenyl phosphate.
Chemical formula	C ₂₄ H ₂₇ O ₄ P
Chemical family	Aryl phosphate
Type of product and use	Fire-resistant hydraulic fluid
Company	ICL-IP America Inc. 622 Emerson Road - Suite 500 St Louis, Missouri 63141, USA Phone - 1 800 666 1200
Emergency telephone number:	
- For Europe	+32 (0) 3 575 5555 SGS
- For USA	Chemtrec (800) 424-9300
- For Asia - Pacific	ALERT-SGS 24 hr Toll Free Number : +800 ALERTSGS (+800-2537-8747) 24 hr Singapore Exchange Number : +65 6542-9595 Tel: 03-3552-1611 Fax: 03-3552-1616
- For Japan	

2. Hazards identification

GHS classification



Signal Word DANGER

Hazard statements
H360F - May damage fertility
H373 - May cause damage to organs through prolonged or repeated exposure
H410 - Very toxic to aquatic life with long lasting effects

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Product name	FYRQUEL EHC-N	
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Precautionary statements	P202 - Do not handle until all safety precautions have been read and understood P260 - Do not breathe dust/fume/gas/mist/vapors/spray P273 - Avoid release to the environment P280 - Wear protective gloves/protective clothing/eye protection/face protection P308 + P313 - IF exposed or concerned: Get medical advice/attention P391 - Collect spillage P405 - Store locked up P501 - Dispose of contents/container in accordance with national and international regulations
--------------------------	--

NFPA Ratings (Scale 0-4) Health = 1, Fire = 1, Reactivity = 0

3. Composition / information on ingredients

Components	Weight %	Index No.	EC No.	EU Classification
Trixylyl phosphate 25155-23-1	100	015-201-00-9	246-677-8	Repr. 1B, H360F STOT RE, H373 Aquatic Acute 1, H400 Aquatic Chronic 1, H410 (In accordance with CLP 1272/2008) Repr. 2; R60 Xn; R48/22 N; R50/53 (In accordance with DSD 67/548/EEC)

4. First-aid measures

Eye contact	Holding the eyelids apart, flush eyes promptly with copious flowing water for at least 20 minutes. Get medical attention immediately.
Skin contact	Remove contaminated clothing. Wash skin thoroughly with mild soap and plenty of water for at least 15 minutes. Wash clothing before reuse. Get medical attention if irritation occurs.
Inhalation	In case of inhalation, remove person to fresh air. Keep him quiet and warm. Apply artificial respiration if necessary and get medical attention immediately.

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Ingestion If swallowed, wash mouth thoroughly with plenty of water. Get medical attention immediately

NOTE: Never give an unconscious person anything to drink

Most important symptoms and effects, acute or delayed

May damage fertility
May cause damage to organs through prolonged or repeated exposure.

Notes to the physician Treat symptomatically and supportively.

5. Fire - fighting measures

Suitable extinguishing media	Material is not combustible. Use extinguishing media appropriate to surrounding fire conditions.
Unusual fire and explosion hazards	When heated to decomposition, may release poisonous and corrosive fumes of carbon dioxide, carbon monoxide and phosphorus oxides.
Fire fighting procedure	Fire fighters should wear full protective clothing and self-contained breathing apparatus (SCBA). Contain runoff to prevent entry into water or drainage systems.

6. Accidental release measures

Personal precautions	Wear appropriate safety clothing and eye/face protection (see Section 8).
Methods for cleaning up	Ventilate area and wash spill site after material pickup is complete. Soak up with sand or other suitable absorbant and dispose of as solid waste. Collect in suitable and properly labeled containers.
Environmental precautions	Prevent product from entering drains, ditches and rivers.

7. Handling and storage

Handling	Avoid bodily contact. Keep containers tightly closed.
Storage	Store in a dry, cool, well-ventilated area away from incompatible materials (see "materials to avoid"). Maximum recommended storage temperature of 65°C (149°F) Store above 4.4°C (40°F) for improved pumping rates. Temperatures between 27 - 37.8°C (80 - 100°F) provide good flow rates

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8. Exposure controls / personal protection

Exposure Limits :

Components	ACGIH-TLV Data	OSHA (PEL) Data
Trixylyl phosphate 25155-23-1	Not determined	Not determined

Ventilation requirements Provide adequate ventilation.

Personal protective equipment:

- **Respiratory protection** Under normal use conditions, no breathing protection is required, due to the nonvolatile nature of the fluid. Full-face fitted air-purifying respirator equipped with organic vapor cartridges is needed when an aerosol is formed with no elevated temperatures. Acid cartridges should be added where the fluid comes into contact with a heated surface with generation of smoke and aerosols.
- **Hand protection** Neoprene gloves
- **Eye protection** Chemical safety goggles
- **Skin and body protection** Body covering clothes and boots.

Hygiene measures Do not eat, smoke or drink where material is handled, processed or stored. Wash hands thoroughly after handling and before eating or smoking. Safety shower and eye bath should be provided.

9. Physical and chemical properties

Appearance Clear, transparent liquid
Odor Slight
Melting point/range -20 - -15°C
Boiling point/range 394.1°C
Pour point -17.77°C (0 °F)
Flash point >200°C
Vapor pressure 0.21 kPa (20°C)
0.27 kPa (25°C)
1.142 (20°C)
Relative density
Solubility:
- **Solubility in water** <0.02 mg/L
Partition coefficient Log Pow = >6.2
(n-octanol/water)
Auto-ignition temperature Not self-ignitable
Viscosity 217.5 mPa*sec dynamic (20°C)
66.6mPa*sec dynamic (35°C)
Explosive properties There are no chemical groups associated with explosive properties present in the molecule

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Oxidising properties No oxidising properties can be predicted on the basis of chemical structure

10. Stability and reactivity

Stability	Stable under normal conditions
Conditions to avoid	Exposure to moisture.
Materials to avoid	Strong oxidizers, strong acids and strong alkalis. It hydrolyzes slowly at normal temperatures in acidic or alkaline aqueous solutions.
Hazardous decomposition products	Carbon dioxide and carbon monoxide Phosphorus oxides,

11. Toxicological information

Acute toxicity:	
- Rat oral LD50	> 5000 mg/kg
- Rabbit dermal LD50	> 2000 mg/kg
- Dermal irritation (rabbit)	Mild irritant.
- Eye irritation (rabbit)	Mild irritant
Sub-chronic toxicity:	Rats that received daily oral doses of 200 mg/kg/day or 1000 mg/kg/day for three or more weeks showed a reduction in plasma cholinesterase activity. Neurotoxicity tests conducted during this subchronic toxicity study, including a functional observation battery and motor activity assessment did not show any signs of neurotoxicity.
Chronic toxicity	No data available
Mutagenicity	Not mutagenic by the Ames Test. Not clastogenic in chromosome aberration test with Chinese hamster ovary cells.
Target organ effects	Combined Oral Repeated Dose and Reproductive/Developmental Toxicity Screening study in rats: affected organs: adrenals, testes, epididymides, ovaries, liver (females only)
Carcinogenicity	Not included in NTP 12th Report on Carcinogens Not classified by IARC

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Reproductive toxicity In a reproductive toxicity test, male and female rats received daily oral doses of the product for two weeks after which they were mated. Although mating was successful, there was a treatment-related decrease in the pregnancies in the mid and high dose animals. While there were no pregnancies in the high dose animals, when the previously treated animals were maintained without treatment for four weeks and then mated, all the females became pregnant. Thus the antifertility effect is fully reversible.

Neurotoxicity Can cause acute delayed neurotoxicity when administered orally to hens at > 1000 mg/kg

12. Ecological information

- 96 Hour-LC50, Fish > 1119 ug/l (Fathead minnow)
> 100 mg/l (Rainbow Trout)

- 48 Hour-EC50, Daphnia magna 60 ug/l

- EC50, Freshwater algae > 1011 ug/l (96 hr)
Biodegradation Not readily biodegradable.

13. Disposal considerations

Waste disposal Liquid material should be incinerated. Material absorbed onto sand or earth should be disposed of as solid waste in accordance with local regulations.

Disposal of Packaging Empty packaging may contain product residues and due consideration should be given prior to disposal

14. Transportation information

UN No. 3082

IMDG Proper shipping name: Environmentally hazardous substance, liquid, n.o.s (trixylenyl phosphate)
Class: 9 - Miscellaneous Dangerous Substances and Articles
Label: 9
Packing Group: III
Mark: MARINE POLLUTANT

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ADR/RID Proper shipping name: Environmentally hazardous substance, liquid, n.o.s (trixylenyl phosphate)
Classification Code: M6
Class: 9 - Miscellaneous Dangerous Substances and Articles
Danger Label Model No.: 9
Hazard identification No: 90
Packing group: III
Marking: Environmentally hazardous substance

ICAO/IATA Proper shipping name: Environmentally hazardous substance, liquid, n.o.s (trixylenyl phosphate)
Class: 9
Hazard label(s): Miscellaneous
Packing group: III
Marking: Environmentally hazardous substance

DOT Proper shipping name: Environmentally hazardous substances, liquid, n.o.s (trixylenyl phosphate)
Class: 9 - Miscellaneous Hazardous Material
Label: 9
Packing Group: III

Not regulated for surface and air transport in non-bulk (<119 gallons) packagings. (contains trixylenyl phosphate which is a Marine Pollutant per 49CFR 172.101 Appendix B)

15. Regulatory information

EU	Reported in EINECS Regulated in France - Possible work-related illnesses, Tables A and B (Code de la Sécurité Sociale, Section III, Annex A, Article D.461-1, July 1999)
USA	Reported in the EPA TSCA Inventory.
Australia	Listed in AICS
Canada	Listed in DSL
CHINA - China inventory	Listed in IECSC
Japan	Listed in ENCS
Korea	Listed in ECL

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Philippines Listed in PICCS

16. Other information

This data sheet contains changes from the previous version in section(s)
2, 3 (REACH), 4, 8, 9, 13, 15

Health, Safety & Environment Policy

We will strive to ensure that our operations and products meet the needs of the present global community without compromising the ability of future generations to meet their needs. We accept that the success of our business is dependent on the supply of products and services that will benefit society whilst ensuring human safety and protection of the environment and natural resources. Within the framework of our commitment to the Responsible Care program, we will provide a healthy and safe work environment for employees and will responsibly manage our products at all stages of their life cycle in order to protect human health and the environment whilst maintaining high production standards of operation.

TO MEET THIS COMMITMENT WE WILL: Comply with or exceed applicable national and international regulatory requirements and other requirements to which we subscribe. Communicate openly and actively encourage dialogue with employees, customers and community concerning our products and operations. Implement documented management systems consistent with and for promotion of the Responsible Care ethics.

Develop and supply products that can be manufactured, transported, used and disposed of safely whilst best meeting the needs of our customers. Regularly assess, continually improve and responsibly manage health, safety and environmental risks associated with products and processes throughout their life-cycles. Share knowledge and expertise with others and seek to learn from and incorporate improved practices into our own operations.

Educate and train employees, contractors and customers to improve their HSE performance. Communicate up-to-date information to enable our workers, customers and other interested parties to handle our products in a safe and environmentally responsible manner. Endeavor to work with customers, suppliers, distributors and contractors to foster the safe use, transport and disposal of our chemicals. Support Product Stewardship programs in cooperation with customers, distributors and transporters.

Although the information and recommendations set forth herein (hereinafter "information") are presented in good faith and believed to be correct as of the date hereof, ICL-IP America Inc. makes no representations as to the completeness or accuracy thereof. Information is supplied upon the condition that the persons receiving same will make their own determination as to its safety and suitability for their purposes prior to use. In no event will ICL-IP America Inc. be responsible for damages of any nature whatsoever resulting from the use of or reliance upon information. NO REPRESENTATIONS OR WARRANTIES, EITHER EXPRESSED OR IMPLIED, OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE OR OF ANY OTHER NATURE, ARE MADE HEREUNDER WITH RESPECT TO INFORMATION OR THE PRODUCT TO WHICH THE INFORMATION REFERS.

In an event of discrepancy between the contents of this SDS and the English version of it, the English version shall prevail.

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Prepared by	HERA Division in ISRAEL telephone: +/972-8-6297835 telefax: +/972-8-6297832 www.icl-ip.com
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End of safety data sheet